

# DR. NIRAJ NAGRALE

## Assistant Professor



### About Me

An experienced academician in Electronics and Telecommunication Engineering, specializing in UG and PG teaching with strong theoretical and practical expertise. Actively mentors students in research by integrating AI, Machine Learning, and Digital Signal Processing with core engineering concepts.

### Contact

+91 9322502352

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Flat No. 302, Mahindra Kasturi III Apartment, Rajendra Nagar, Hingna Road, Nagpur 440036 India

### Publication

0000-0002-1344-5522

58365328800

• Web of Science Indexed : 04  
Journal Articles

• International Conference : 12  
(Scopus/WoS Indexed)

• Book : 01

• Book Chapter : 02

• Total Publications : 27

### Patent/Copyright

• Granted : 01

• Published : 05

• Copyright(Registered) : 04

### Reserch Area

Digital Image Processing IoT DSP AI  
Machine Learning Soft Computing

### Experience

27 Years

#### Assistant Professor

G H R C E, Nagpur

2022 – 2026

- Teach theory and laboratory courses to UG and PG students in Electronics and Telecommunication Engineering.
- Supervise 100+ minor and major student projects.
- Contribute to question paper setting, syllabus design, and OBE-aligned curriculum development.
- Serve as **Coordinator of the IETE** professional body, promoting technical activities, workshops, and industry-academia interaction.
- Mentor UG and PG students in research leading to joint research publications.

#### Assistant Professor

R C E R T, Chandrapur

1998– 2022

- Teach theory and laboratory courses to UG and PG students in Electronics and Telecommunication Engineering.
- Supervise 80+ minor and major student projects.
- Contribute to question paper setting, syllabus design, and OBE-aligned curriculum development.
- Serve as **IEEE Counselor**, promoting student membership, research, and professional activities.
- Serve as **Dean – Student Affairs**, overseeing student welfare and academic coordination.

### Education

Bharti Vishwavidyalaya, Durg

PhD

RGPV, Bhopal

Master of Technology (MTech) in ECE

RTMNU, Nagpur

Master of Technology (MTech) in EMS

RTMNU, Nagpur

Bachelor of Engineering(BE) in Electronics Engineering

# Dr. Niraj Kesharao Nagrale



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Mahindra Kasturi III Apartment,  
Behind Vasudev Nagar Metro Station,  
Rajendra Nagar, Hingna Road,  
Nagpur 440036 (M.S.)

**Date of Birth** : 01 June 1973

**Gender** : male

**Nationality** : Indian

**Marital Status** : Married

## Academic Qualification

| Degree/<br>Examination | College/ Institute                                                           | Affiliated<br>University/<br>Board/<br>Autonomous<br>Body                               | Passing<br>Year | Discipline                                             | Division/<br>Class | CGPA/<br>%es |
|------------------------|------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------|--------------------|--------------|
| PhD                    | Bharti Vishwavidyalaya,<br>Durg.                                             | Bharti<br>Vishwavidyalaya<br>Durg                                                       | 2025            | Electronics &<br>Tele-<br>Communication<br>Engineering | ---                | ----         |
| MTech.                 | Shree Balaji Institute of<br>Technology & Management,<br>Betul               | RGPV, Bhopal                                                                            | 2020            | Electronics &<br>Communicati<br>on<br>Engineering      | First              | 7.85<br>CGPA |
| MTech.                 | Rajiv Gandhi College of<br>Engineering, Research &<br>Technology, Chandrapur | RTM Nagpur<br>University<br>Nagpur                                                      | 2005            | Energy<br>Management<br>System                         | First              | 75.59%       |
| BE                     | Rajiv Gandhi College of<br>Engineering, Research &<br>Technology, Chandrapur | RTM Nagpur<br>University<br>Nagpur                                                      | 1997            | Electronics<br>Engineering                             | First              | 60.00%       |
| HSSC                   | Sent Michael Junior College<br>of Science, Chandrapur                        | The Maharashtra<br>State Board of<br>Secondary and<br>Higher<br>Secondary<br>Education, | 1991            | SCIENCE                                                | Second             | 57.50%       |

## NPTEL Online Certification:

|   |                                                                |                              |                |               |               |             |
|---|----------------------------------------------------------------|------------------------------|----------------|---------------|---------------|-------------|
| 1 | Introduction to Soft Computing                                 | Jan-Mar 2025                 | 8-week course  | IIT Kharagpur | Elite         | 3-4 Credits |
| 2 | Introduction to IoT                                            | 22 July 2024 to 11 Oct. 2024 | 12-week course | IIT Kharagpur | Elite+ Silver | 3-4 Credits |
| 3 | Introduction To Industry 4.0 And Industrial Internet of Things | Jan- Apr 2024                | 12-week course | IIT Kharagpur | Elite         | 3-4 Credits |
| 4 | Introduction to Machine Learning                               | Jul-Sep 2023                 | 8-week course  | IIT Kharagpur | Elite         | 2 Credit    |

## Experience Details (Latest First):

| Sr. No | Organization                                                                          | Designation                 | Date       |            | Total Period             |
|--------|---------------------------------------------------------------------------------------|-----------------------------|------------|------------|--------------------------|
|        |                                                                                       |                             | From       | To         |                          |
| 1      | G H Raison College of Engineering, CRPF Gate, No.3, Hingna Road, Digdoh Hills, Nagpur | Assistant Professor         | 10-03-2022 | Till Date  | 4 Years                  |
| 2      | Rajiv Gandhi College of Engineering, Research and Technology, Chandrapur              | Assistant Professor Grade 3 | 22-01-2017 | 09-03-2022 | 5 years, 2 Months        |
|        |                                                                                       | Assistant Professor Grade 2 | 22-01-2012 | 21-01-2017 | 5 years                  |
|        |                                                                                       | Assistant Professor         | 26-04-2006 | 21-01-2012 | 5 years 8 months 26 days |
|        |                                                                                       | Lecturer                    | 01-07-2005 | 25-04-2006 | 9 months 25 days         |
|        |                                                                                       | Lecturer                    | 10-07-1998 | 30-04-2005 | 7 years 9 months 22 days |

## PUBLICATIONS AND PATENT

### Patent

| Sr No. | Title Of the Patent                                                                  | Patent Application Number | Date of Filing | Status    |
|--------|--------------------------------------------------------------------------------------|---------------------------|----------------|-----------|
| 1      | Indian Design Patent<br>AI BASED SOIL TESTING DEVICE                                 | 438196-001                | 25/12/2024     | Granted   |
| 2      | Advanced Ear biometrics Authentication Using EfficientNet CNN for Enhanced Security. | 202421069667              | 14/09/2024     | Published |
| 3      | A System for Ear Biometric authentication                                            | 202421011538              | 19/02/2024     | Published |
| 4      | A Rocket Landing System                                                              | 202421016952              | 08/03/2024     | Published |
| 5      | A Power Distribution Control System                                                  | 202421018359              | 13/03/2024     | Published |

## Book

|                                                                                                |                               |                             |
|------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------|
| <b>Title:</b> Crowd analysis – Using deep neural networks to analyze images from CCTV cameras. |                               |                             |
| <b>Publisher:</b> LAP LAMBERT Academic Publishing.                                             |                               | <b>ISBN:</b> 9786208010355. |
| Number of pages: 184                                                                           | <b>Book language:</b> English | Price: (EUR) €79.90         |

## International Journals

| Sr. No. | Title of the article                                                                                   | Name of the Journal                                                                                                                  | Publisher      | Year | Indexing                             | Link                                                                                                                                                                                                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|----------------|------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Optimized and Challenges in Ear Biometrics for Secure Identification                                   | Advanced International Journal of Multidisciplinary Research<br>Volume 4, Issue 1, January-February 2026                             | AIJMR          | 2026 | Web of Science (WOS), Google scholar | <a href="https://www.aijmr.com/research-paper.php?id=1203">https://www.aijmr.com/research-paper.php?id=1203</a>                                                                                                                                                                             |
| 2       | Optimized Ear Biometric Recognition Using EfficientNet and PCA: A Deep Learning Approach               | International Journal on Science and Technology (IJSAT) E-ISSN: 2229-7677, Volume 16, Issue 1                                        | IJSAT          | 2025 | Web of Science (WOS), Google scholar | <a href="https://www.ijstat.org/research-paper.php?id=2191">https://www.ijstat.org/research-paper.php?id=2191</a><br><a href="https://doi.org/10.71097/IJSAT.v16.i1.2191">doi.org/10.71097/IJSAT.v16.i1.2191</a>                                                                            |
| 3       | Deep Learning Approaches for Ear Biometrics: A Novel Approach                                          | International Journal of Scientific Research in Engineering and Management (IJSREM), Volume 08, Issue 07, July 2024, ISSN: 2582-3930 | IJSREM         | 2024 | Web of Science (WOS), Google scholar | <a href="https://ijsrem.com/download/deep-learning-approaches-for-ear-biometrics-a-novel-approach/">https://ijsrem.com/download/deep-learning-approaches-for-ear-biometrics-a-novel-approach/</a><br><a href="https://doi.org/10.55041/IJSREM36217">doi.org/10.55041/IJSREM36217</a>        |
| 4       | Collaborative Landscape and Knowledge Dynamics in IoT and Blockchain Research: A Bibliometric Analysis | International Journal on Recent and Innovation Trends in Computing and Communication ISSN: 2321-8169 Volume: 11 Issue: 9, 2023       | IJRITCC        | 2023 | Web of Science (WOS), Google scholar | <a href="https://ijritcc.org/index.php/ijritcc/article/view/8988">https://ijritcc.org/index.php/ijritcc/article/view/8988</a><br><a href="https://doi.org/10.17762/ijritcc.v11i9.8988">doi.org/10.17762/ijritcc.v11i9.8988</a>                                                              |
| 5       | Fast algorithm for underwater video compression                                                        | International Journal of Electronics Engineering (ISSN No. 0973-7383), Vol.-II, No. 2 of 2010.                                       | csjournals     | 2010 | Google Scholar                       | <a href="https://csjournals.com/IJEE/PDF%202-2/Article_5.pdf">https://csjournals.com/IJEE/PDF%202-2/Article_5.pdf</a>                                                                                                                                                                       |
| 6       | Satellite image segmentation, soft computing techniques: A survey                                      | TECH-Chronicle An International E-journal on Emerging trends in science, technology and management ISSN No. 2454-1958, PP 137-141    | TECH-Chronicle | 2016 | --                                   | <a href="https://www.researchgate.net/publication/381670810_Satellite_Image_Segmentation_Soft_Computing_Techniques_A_Survey#fullTextFileContent">https://www.researchgate.net/publication/381670810_Satellite_Image_Segmentation_Soft_Computing_Techniques_A_Survey#fullTextFileContent</a> |
| 7       | ENHANCEMENT OF CLOUD DATA SECURITY USING WATERMARKING TECHNIQUE                                        | International Journal of Scientific Research in Engineering and Management (IJSREM), Volume: 04 Issue: 07 July 2020, ISSN: 2582-3930 | IJSREM         | 2020 | --                                   | <a href="https://ijsrem.com/download/enhancement-of-cloud-data-security-using-watermarking-technique/">https://ijsrem.com/download/enhancement-of-cloud-data-security-using-watermarking-technique/</a>                                                                                     |
| 8       | AI Business Boost Approach for Small Business and Shopkeepers: Advanced Approach for Business          | <i>Book chapter</i> in: Digital Twin Technology and AI Implementations in Future-Focused Businesses                                  | IGI Global     | 2024 | Scopus                               | <a href="https://www.igi-global.com/chapter/ai-business-boost-approach-for-small-business-and-shopkeepers/336448">https://www.igi-global.com/chapter/ai-business-boost-approach-for-small-business-and-shopkeepers/336448</a>                                                               |

## International Conference

| Sr. No. | Title of the article                                                                   | Name of the Conference                                                                                                                           | Publisher                    | Year | Indexing       | Link                                                                                                                                                                                                                                                      |
|---------|----------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|------|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | Smart Glove: A Sensor-Driven Communication Aid for the Hearing and Speech Impaired     | 025 IEEE International Conference on Intelligent Signal Processing and Effective Communication Technologies (INSPECT)                            | IEEE                         | 2026 | Scopus         | <a href="https://ieeexplore.ieee.org/abstract/document/11350887">https://ieeexplore.ieee.org/abstract/document/11350887</a>                                                                                                                               |
| 2       | ECG Denoising: From Wavelet Transform to Artificial Intelligence-A Comparative Study   | 2025 12th International Conference on Emerging Trends in Engineering & Technology-Signal and Information Processing (ICETET-SIP)                 | IEEE                         | 2025 | Scopus         | <a href="https://ieeexplore.ieee.org/abstract/document/11156810">https://ieeexplore.ieee.org/abstract/document/11156810</a>                                                                                                                               |
| 3       | Track Sensing Obstacle Avoidance Technology for Autonomous Vehicle                     | 10th IEEE International Students Conference SCEECS 2025, MANIT, Bhopal                                                                           | IEEE                         | 2025 | Scopus         | <a href="https://ieeexplore.ieee.org/abstract/document/10940470">https://ieeexplore.ieee.org/abstract/document/10940470</a>                                                                                                                               |
| 4       | IoT Based Smart Food Grain Warehouse                                                   | 2nd International Conference on Paradigm Shifts in Communications Embedded Systems, Machine Learning and Signal Processing (PCEMS), VNIT, Nagpur | IEEE                         | 2023 | Scopus         | <a href="https://ieeexplore.ieee.org/abstract/document/10136049">https://ieeexplore.ieee.org/abstract/document/10136049</a>                                                                                                                               |
| 5       | A Novel Voice-Activated Cleaning System for Smart Homes                                | 10th IEEE International Students Conference SCEECS 2025, MANIT, Bhopal                                                                           | IEEE                         | 2025 | Scopus         | <a href="https://ieeexplore.ieee.org/abstract/document/10940209">https://ieeexplore.ieee.org/abstract/document/10940209</a>                                                                                                                               |
| 6       | Modern Food Grain Storage with RFID Security Cover and Innovative Technology Using IoT | International Conference on Innovations and Challenges in Emerging Technologies (ICICET)                                                         | IEEE                         | 2024 | Scopus         | <a href="https://ieeexplore.ieee.org/abstract/document/10616361">https://ieeexplore.ieee.org/abstract/document/10616361</a>                                                                                                                               |
| 7       | Implementation of auto serve pro-based management system using python                  | 1st International Conference on Design and Materials (ICDM-2021) Delhi Technological University (DTU), Delhi, India                              | AIP Publishing               | 2025 | Scopus         | <a href="https://pubs.aip.org/aip/acp/article-abstract/3233/1/020036/3331638/Implementation-of-auto-serve-pro-based-management">https://pubs.aip.org/aip/acp/article-abstract/3233/1/020036/3331638/Implementation-of-auto-serve-pro-based-management</a> |
| 8       | Advanced underwater video compression algorithm                                        | International Conference on Computer Applications                                                                                                | Research Publishing Services | 2010 | Google Scholar | <a href="https://rpsonline.com.sg/proceedings/9789810873042/html/978-981-08-7304-2_1431.xml">https://rpsonline.com.sg/proceedings/9789810873042/html/978-981-08-7304-2_1431.xml</a>                                                                       |

| Sr. No | Title of the article                                                                       | Name of the Conference                                                                                                                  | Publisher      | Year | Indexing | Link                                                                                                                                        |
|--------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|----------------|------|----------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 9      | Enhancing Ear Detection for Biometric Systems Using YOLOv9                                 | 3rd International Conference on Paradigm Shifts in Communication, Embedded Systems, Machine Learning and Signal Processing (PCEMS 2024) | Spinger Nature | 2025 | Scopus   | <a href="https://link.springer.com/chapter/10.1007/978-3-031-90580-3_24">https://link.springer.com/chapter/10.1007/978-3-031-90580-3_24</a> |
| 10     | Optimization of Resist Performance in Deep UV Lithography for Sub-10nm Feature Fabrication | 2nd International Conference on Numerical Optimization in Engineering and Sciences (NOIEAS - 2024) NIT Warngal.                         | Spinger        | 2026 | Scopus   |                                                                                                                                             |

## Copyright

| Sr. No. | Titled                                                                                   | Diary Number     | Registration Number | Type of Document   |
|---------|------------------------------------------------------------------------------------------|------------------|---------------------|--------------------|
| 1       | Impedance Matching Devices                                                               | 21281/2022-CO/L  | L-122233/2023       | Literary/ Dramatic |
| 2       | Solved Problems on The Topic Transmission Line                                           | 22012/2023-CO/L  | L-134537/2023       | Literary/ Dramatic |
| 3       | Workflow of Design of An Efficient Ear Biometrics Model using Deep Learning Architecture | 35451/2023-CO/L  | L-151046/2024       | Literary/ Dramatic |
| 4       | Handwritten notes on Guided waves and waveguides                                         | 38467/2024-CO/L  | L-161862/2025       | Literary/ Dramatic |
| 5       | Magnetic Field Calculations Using Biot–Savart’s Law: A Practical Approach                | LD-24893/2025-CO | LD-2025017421       | Literary/ Dramatic |

## Reviewers

| Sr. No | Reviewers of the articles of: | Details of Book/ Conference/Journal                                                                                                        | Type of article | year |
|--------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
| 1      | Book                          | IGI Global Book: Harnessing AI and Digital Twin Technologies in Businesses.                                                                | Book Chapter    |      |
| 2      | IEEE International conference | 11 <sup>th</sup> International Conference on Emerging Trends in Engineering & Technology Signal and Information Processing (ICETET SIP-23) | Research Papers | 2023 |
| 3      | IEEE International conference | International Conference on Innovations and Challenges in Emerging Technologies 2024 (ICICET-2024)                                         | Research Papers | 2024 |
|        | IEEE International conference | IEEE CONFERENCE ON COMPUTING APPLICATIONS AND SYSTEMS (COMPAS 2024)                                                                        | Research Papers | 2024 |
| 4      | IEEE International conference | YCCE Nagpur                                                                                                                                | Research Papers | 2025 |
| 5      | IEEE International conference | International Conference on Sustainability, Innovation & Technology (ICSIT 2025) Symbiosis, Nagpur                                         | Research Papers | 2025 |
| 6      | IEEE International conference | 2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)                                   | Research Papers | 2025 |

| Sr. No | Reviewers of the articles of: | Details of Book/ Conference/Journal                                                                                                        | Type of article | year |
|--------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------|
| 7      | IEEE International conference | 12 <sup>th</sup> International Conference on Emerging Trends in Engineering & Technology Signal and Information Processing (ICETET SIP-25) | Research Papers | 2025 |
| 8      | IEEE International conference | 3rd International Conference on Emerging Trends in Engineering and Medical Sciences, YCCE Nagpur                                           | Research Papers | 2026 |

## Memberships of Professional Bodies:

1. Associate Member, **The Institution of Engineers (India)**, AM094435-4
2. Life Member, **Indian Society of Technical Education**, New Delhi, LM081251

## Development of Course Structure/Syllabi

1. Prepared course structure and syllabi for **B.E. (Electronics Engineering** and related specializations) affiliated to RTMNU as Member of workshop on Curriculum Development for Electronics/ Electronics and Telecommunication/ Electronics Communication Engineering”, held on 5<sup>th</sup> December 2009, organised by S. B. Jain Institute of Technology, Management and Research, Nagpur in collaboration with Rashtra Sant Tukdoji Maharaj Nagpur University, Nagpur.
2. Prepared course structure and syllabi for **B.E. (Electronics Engineering** and related specializations) affiliated to Gondwana University, Gadchiroli as Member of workshop during July 2012.

## Responsibilities/Academic Works for RTMNU, Nagpur and Gondwana University, Gadchiroli

1. Chief Supervisor, of Gondwana University, Gadchiroli to conduct Theory Examination.
2. External Supervisor of Nagpur University, Nagpur to conduct Theory Examination.
3. External Supervisor of Gondwana University, Gadchiroli to conduct Theory Examination.
4. Supervisor of M.Tech.(EMS) project.
5. Supervisor of M.Tech.(CME) project.

## Responsibilities/Academic Work of the College:

1. Dean, Student Affairs.
2. Counselor of IETE
3. Counselor of IEEE Student Branch, RCERT, Chandrapur
4. Organising committee member of All India Marathi Literary Conference hosted by college.
5. Coordinator of “Crest-o-mania” State level paper presentation and model exhibition competition held on 20-21<sup>st</sup> February 2009, organised by ISTE Student Chapter, RCERT, Chandrapur.
6. Coordinator of ISTE Student Chapter RCERT, Chandrapur (July 2000 to May 2010).
7. ONE WEEK ONLINE SHORT-TERM TRAINING PROGRAM (STTP) ON Artificial Intelligence for Solving Emerging Industry Challenges: Trends, Tools and Application.



**Dr. Niraj Keshaorao Nagrale**